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Autism Spectrum Disorder Detection using Navie Bayes Tree Technique

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ABSTRACT:

Autism Spectrum Disorder (ASD) impacts the social and communicative abilities of individuals of all ages. Autism screening is a time-consuming and complicated procedure, making it difficult to discover autistic individuals. Recently, machine learning techniques have been applied to streamline the screening process and rapidly discover ASD. However, it is necessary to increase the accuracy of predictions using the limited data available. This research provides a novel approach for detecting ASD using the Nave Bayes (NB) Tree. The proposed method generates a decision tree from a data set using the naive Bayes categorizer at the nodes to assess whether or not a person has autism. The performance of the suggested technique is compared to that of the most commonly used classification algorithms, including Nave Bayes, Support Vector Machine (SVM), and Multilayer Perceptron (MLP). The results of the experiments show that the NB Tree-based technique is better than the current classification methods in terms of classification accuracy, Root Mean Squared Error (RMSE), and precision.

Keywords: *Autism, Autism Screening, Machine Learning, Naïve Bayes Tree, ASD*

I.INTRODUCTION:

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that affects communication, repetitive and restricted behaviors (RRBs), and social behavior of individuals of all ages. According to the World Health Organization, one in 160 children worldwide have autism spectrum disorder (3). It affects one of the most important brain areas, the cerebellum, which is crucial for motor abilities (Stone W, Coonrod E and Ousley O. 2000). It is discovered that ASD has several causes, including environmental and genetic aspects. Since there is no medically established therapy for ASD, it is crucial to detect the disorder sooner so that psychosocial therapies may be initiated.

The Autism Diagnostic Interview (ADI) and Autism Diagnostic Observation Schedule (ADOS) are often used to diagnose ASD (Stone W, Coonrod E and Ousley O. 2000, Lord C, Risi S and Lambrecht L.

2000). The procedure of diagnosing ASD is often difficult and time-consuming. The present screening approach in the United Kingdom takes three years to identify ASD (Fadi Thabtah and David Peebles. 2020). Methods such as Screening Tool for Autism in Toddlers and Young Children (STAT), Childhood Autism Rating Scale (CARS-2) and Autism Spectrum Quotient (AQ) have been developed and used in various nations to reduce the time required for diagnosis (Stone W, Coonrod E and Ousley O. 2000, Schopler E and Bourgondien M. 2010, Schopler E and Bourgondien M. 2010, Baron-Cohen S, Wheelwright S, Skinner R. 2001). Existing screening approaches must examine a large number of behavioural characteristics to determine if an individual has ASD. Most existing screening approaches, such as STAT and CARS-2, use a questionnaire to analyse the respondents' behaviour. The test score is calculated based on the responses to the questions and the rules used to identify ASD.

In recent years, machine learning techniques have been used to diagnose ASD for two important reasons (Fadi Thabtah and David Peebles. 2020). They have accelerated the diagnosis procedure by reducing the number of symptoms to be examined during the screening test (Geschwind DH, Sowinski J, Lord C. 2001, Lord C, Risi S and Lambrecht L. 2000). They also boost the diagnostic procedure's sensitivity and specificity. In the literature, Autism Spectrum Disorder has been classified in the literature using machine learning approaches such as Decision Trees, Neural Networks, Support Vector Machines, and Bayesian Networks (Fadi Thabtah. 2019). However, they have used distinct datasets for various age groups. Consequently, the categorization approach that works well for children is unsuitable for adolescents and adults. In addition, the majority of extant research does not test the model on all three age groups: children, adolescents, and adults.

Decision tree methods are widely used for classification jobs in a variety of disciplines (Quinlan J. 1986). In contrast to other classification models, the output of a decision tree is readily interpretable as if-then rules. In addition, decision trees are more transparent than other models since they analyse all conceivable choices and trace the consequences of each alternative to the conclusion in a single perspective. Therefore, it may be used by all types of individuals, including non-technical users. In addition, decision trees do not require data standardization because they can accommodate missing values for features. However, it is incapable of handling continuous and missing data, and it is susceptible to overfitting. The Naive Bayes Model performs exceptionally well when the training data does not include all possible outcomes. It has an exceptionally high classification accuracy with a small amount of data. However, it assumes strong independence, which influences its classification accuracy. Compared to Naive Bayes, decision trees perform better with large amounts of data. However, neither the Naive Bayes nor the Decision Tree methods scale well for huge datasets (Ron Kohavi .1996). This article employs the Nave Bayes Tree classifier for ASD diagnosis, which combines the best features of the Decision Tree and Nave Bayes classifiers. It compares the performance of the Nave Bayes tree approach to the performance of the current classification methods utilising three ASD data sets associated with children, adolescents, and adults that have been published in the University of California Irvine (UCI) Repository. The results of the experiment show that the suggested method for classifying is more accurate than the current methods.

The structure of this document is as follows: The second section explains the different machine learning methodologies used to diagnose ASD, while the third section describes the tree-based classification techniques used to identify ASD. The fourth section examines the empirical findings, while the fifth section ends the study.

II. LITERATURE SURVEY:

The Standard Clinical Judgment (CJ) approach is now used by specialist doctors to diagnose ASD. The Diagnostic and Statistical Manual, Third Edition, Revised (DSM-III-R), Autism Behavior Inventory (ABI), Autism Diagnostic Observation Schedule (ADOS), Checklist for Autism in Toddlers (CHAT), and Childhood Autism Rating Scale (CARS) have all been utilised in autism diagnosis (Lopez Marcano JL. 2016, Fadi Thabtah. 2019). Clinicians typically use one or two methods to determine whether a patient has an ASD. During the screening procedure, these methods examine numerous observable and quantifiable behavioural indicators and calculate a score. They use the embedded function's static rules to determine if ASD is present. In recent years, machine learning methods requiring an optimal number of indicators (features) have been applied to the diagnostic test in an effort to reduce diagnostic time. Very few machine learning algorithms have been applied to autism diagnosis in the literature.

Baron-Cohen, Wheelwright, and Skinner (2001) used a Mulitlayer Feed Forward Neural Network to diagnose autism. However, a small sample generated by the Autistic Behavior Interview was utilised (ABI). Florio, T., Einfeld, S., Tonge, B., and Brereton, A. (2009) compared the performance of the Multilayer Perceptron model to that of the Linear Regression model. In contrast to Cohen, they evaluated MLP using the Developmental Behaviour Checklist (DBC) data set. They claimed that the accuracy of neural network predictions for the DBC data set is 80%. Wall DP, Dally R, and Luyster R. (2012) have used numerous machine learning models, including decision tree methods, C4.5 (J48), CART, Nearest Neighbor (NN), and rule-based systems to detect autism (Wall DP, Dally R, Luyster R. 2012, Wall DP, Kosmiscki J, Deluca TF. 2012.). Using the ADOS-Revised-Module 1 diagnostic method, they used a set of data that was not balanced and had some missing information.

The Autism Diagnostic Interview-Revised (ADI-R) dataset in the Autism Genetic Resource Exchange (AGRE) repository is used to diagnose ASD in Wall DP, Dally R, and Luyster R. (2012). Using WEKA, they compared the false positive rate (FPR), the true positive rate (TPR), and the accuracy of these approaches. Based on what they found, the Alternating Decision Tree (ADTree) and the Functional Tree (FT) were more accurate than all the other methods. However, there are conceptual and practical difficulties with these techniques (Bone D, Goodwin MS, Black MP. 2014). According to Wall and colleagues, the ADOS-Revised must be done in a clinical setting and not self-administered. Autism screening has included unsupervised machine learning algorithms like Learning Vector Quantization (LVQ), Self-Organizing Maps (SOM), Fuzzy C-means clustering, and K-means clustering. Using CARS, they've produced datasets and evaluated the severity of autism. The number of cases correctly classified (CCR) for categorizing autistic patients into normal, moderate, and severe categories was greatest for the SOM model.

Chen et al. [1] introduced a fuzzy-established method for predicting temperatures in Taipei. They employed a sliding window approach to forecast short-term temperature trends using a smaller dataset. In contrast, Maqsood et al. [2] utilized an ensemble approach to estimate temperatures in Canada. This method demonstrated the capability to handle larger datasets when compared to fuzzy lucidity.

[3] conducted a comprehensive survey of various machine-learning techniques for weather forecasting. However, these techniques often require extended learning periods when predicting weather conditions. Sawaitul et al. [4] and Naik et al. [5] employed a back propagation procedure to

make endure predictions, with modified wisdom parameters based on the outdated Levenberg–Marquardt algorithm. [6] applied these adjusted wisdom parameters to conjecture weather conditions in Chennai using data from 2010.

Papacharalampous et al. [7,8] merged dissimilar time series copies using four years of climate data to predict one-month climate conditions. They used accidental walks and a naive procedure to segment the 4th year dataset. Techniques like inclinations, autoregressive combined poignant averages (ARIMA), and the clairvoyant method were employed for weather forecasting. The recurrent trend, in combination with the prophet method, proved particularly accurate in weather prediction compared to conventional approaches.

III. PROPOSED APPROACH:

Figure 1 depicts the architecture of the proposed ASD detection model utilizing the Nave Bayes Tree (NB Tree). Using the ASD data set for children, adolescents, and adults, the NB Tree has developed a categorization model for the suggested model. The Nave Bayes Tree (NB Tree) is a classification method that combines the Nave Bayes and Decision Trees (Ron Kohavi, 1996). It removes both the constraints of the Naive Bayes method and the Decision Tree algorithm. It divides the datasets into subgroups using the C4.5 decision tree-like recursive partitioning technique. However, the leaf nodes employ Naive Bayes classifiers to identify the class.

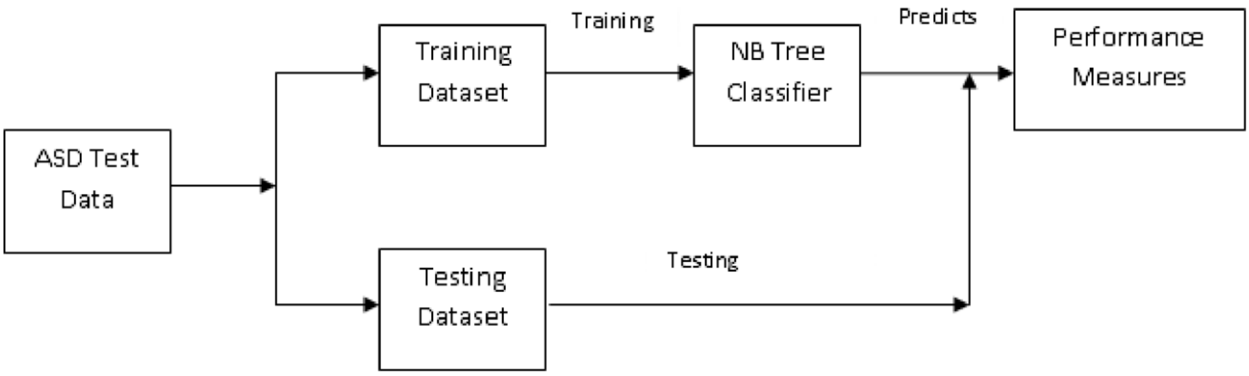

Figure 1: Architecture of the Proposed System

First, the data set is separated into training and testing data sets. The following approach is then used to generate an NB Tree from the training dataset.

Algorithm: NBTree(D)

Input: an attribute – valued dataset D
Tree = {}
for all attribute $a_i \in D$ do
compute utility $u(a_i)$
end for
Select a_j = argmax (ai) // the best attribute with the highest utility
Tree = Create a NB Tree node which tests a_j at the root
if aj is continuous then
D_v = Partition the D according to test on a_j using the threshold-based split
else // aj is discrete,
D_v = Partition the D according to test on a_j using multi-way-based split
end if
for all D_v do
subTree = NBTree(D_v) // Recursive partition
Attach the subTree to the Tree
end for
return Tree

Once the model is built, it is tested using the testing data set. The model performance is then assessed using the standard performance measures. The proposed model is then made available to healthcare professionals for practice.

IV. EXPERIMENT RESULTS:

The performance of the suggested ASD Classification approach is evaluated using Waikato Environment for Knowledge Analysis (WEKA) (Hall M, Frank E, Holmes G. 2009). WEKA is an open-source, Java-based application that is offered under the GNU General Public License. The UCI Repository is the source for ASD datasets for children, adolescents, and adults (24). Table.1 provides a list of the data sets' most important characterise.

Table.1 Input Parameters

Attribute	Type	Description
Age	Integer	Age expressed in years
Gender	String	Male or Female
Ethnicity	String	List of prevalent ethnic groups in text format
Born with jaundice	Boolean (yes or no)	If the patient was born with jaundic
Family member with PDD	Boolean (yes or no)	Whether any member of the immediate family has a PDD
Who is completing the tes	String	Parent, self, caregiver, medical personnel, clinician,etc.
Country of residence	String	Country listing in text format
Question 1 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 2 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 3 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 4 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 5 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 6 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.

Question 7 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 8 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 9 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Question 10 Answer	Binary Value (0 or1)	The question's response code dependent on the screening process used.
Screening Score	Integer	The ultimate score determined by th screening technique's scoring algorithm. This was determined using automated computation.

The classification accuracy of the proposed method is examined by several performance metrics such as Accuracy, Kappa, RMSE, F Measure, Precision, Recall, Methews Correlation, Area under ROC and Area under PRC. The following table compares the results of the NB Tree with various classification algorithms such as Naïve Bayes, Support Vector Machine (SVM), K-Nearest Neighbour (KNN), Multilayer Perceptron (MLP), Hoeffding Tree.

As can be seen in Table.2, compiles a table that tabulates the values of every statistic for each of the six classifiers applied to the adult dataset. The KNN classifier has the worst performance metrics out of the six available options. When compared to the performance of other classifiers, SVM, Hoeffding, and NB Tree all have varying degrees of accuracy and RMSE alone.

Table 2. Performance Evaluation of classifiers on Adult Dataset

Performance Measures	**Naïve Bayes**	**SVM**	**KNN**	**MLP**	**Hoeffding**	**NBTree**
Accuracy	92.84	93.16	88.36	92.69	93.23	97.12
Kappa	0.98	1	0.77	0.99	1	1
RMSE	0.11	0.08	0.33	0.04	0.03	0
F Measure	0.99	1	0.88	1	1	1
Precsion	0.99	1	0.92	1	1	1
Recall	0.99	1	0.85	0.99	1	1
methews Correlation	0.98	1	0.78	0.99	1	1
Area_under_ROC	1	1	0.88	1	1	1
Area_under_PRC	1	1	0.86	1	1	1

As demonstrated in Figure.1, when comparing the accuracy of all six classifiers for the adult dataset, the NB Tree classifier achieves the maximum accuracy. Compared to other classification techniques, KNN's predictive accuracy is worse. It is also noted that NB, SVM, MLP, and Hoeffding algorithms produce more than 90 percent accuracy, however the RMSE, methews Correlation, and Recall are somewhat lower for Hoeffding and NB Tree. Perhaps the NB Tree's prediction accuracy is considerably higher than Hoeffding's.

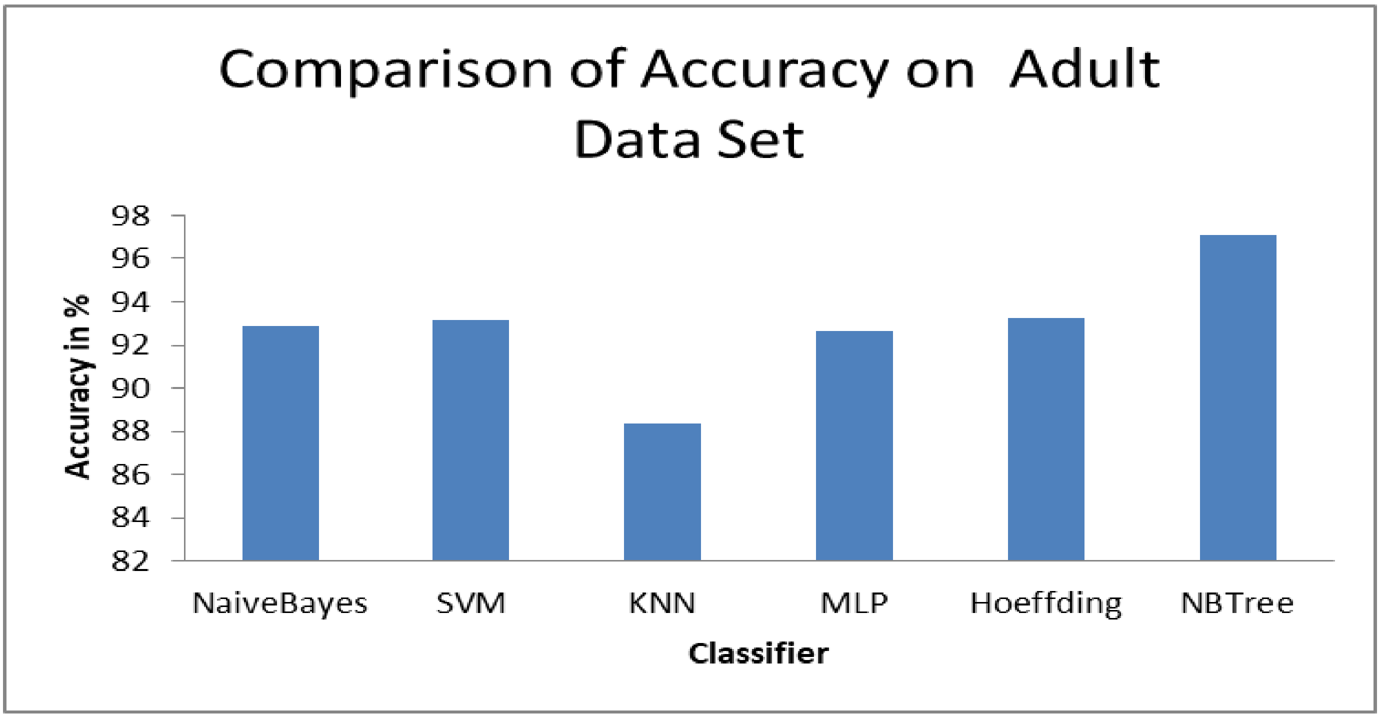

Figure.3 Accuracy of classifiers on Adult Dataset

Table.3 illustrates that NB Tree achieves superior results compared to all other classifiers used in the adolescent dataset. In contrast to the adult dataset, the RMSE, Recall, and Methews Correlation are all considerably improved when utilising the NB Tree instead of the Hoeffding Tree for the analysis of adolescent data. And as is the case with the adult data set, KNN demonstrates a poor performance in terms of performance indicators. Even if SVM MLP and Hoeffding have a higher degree of accuracy compared to KNN, their performance is not as good as NBTree's

Table 3. Performance Evaluation of classifiers on Adolescent Dataset

Performance Measure	**Naïve Bayes**	**SVM**	**KNN**	**MLP**	**Hoeffding**	**NBTree**
Accuracy	91.3	90.77	89.81	90.89	91.97	95.91
Kappa	0.94	0.81	0.77	0.81	0.96	1
RMSE	0.11	0.25	0.26	0.23	0.11	0.02
F Measure	0.96	0.88	0.85	0.88	0.97	1
Precsion	1	0.92	0.97	0.91	1	1

Recall	0.93	0.87	0.77	0.88	0.95	1
methews Correlation	0.95	0.82	0.81	0.82	0.96	1
Area_under_ROC	1	0.9	0.88	0.98	1	1
Area_under_PRC	1	0.85	0.84	0.98	1	1

As can be seen in the figure, the accuracy of the NB, SVM, MLP, and Hoeffding models remains the same when applied to the adolescent data set. However, in comparison to other classifiers, NB Tree has a much greater level of precision and accuracy. In addition, the NB and Hoeffding classifiers both provide the same results in terms of all performance indicators. In a similar vein, the SVM, KNN, and MLP have all shown the same performance in terms of all performance metrics.

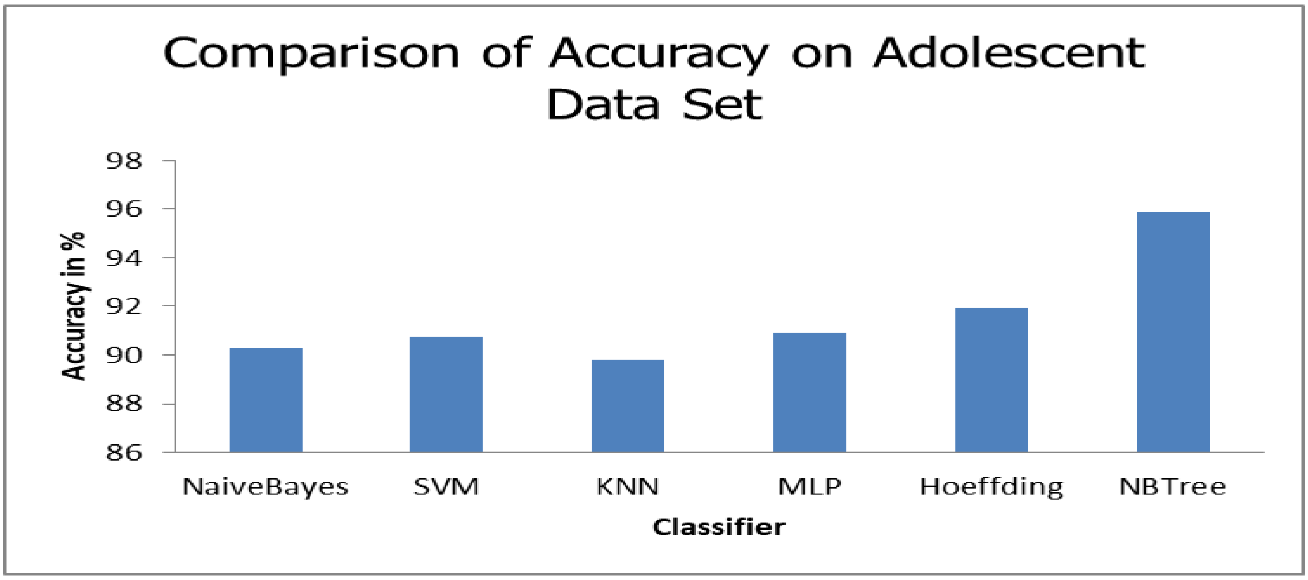

Fig.3 Accuracy of classifiers on Adolescent Dataset

Table.4 demonstrates that NB Tree and Hoeffding Tree have demonstrated the same performance for children data set compared to other classifiers. In contrast to prior datasets, the performance of MLP on the adult dataset is much superior than that of NaiveBaye, SVM, and KNN. The experimental findings demonstrate that regardless of the data set, NB Tree classifies the ASD test data more precisely than any other algorithm.

Table 4. Performance Evaluation of Classifiers on Children dataset

Performance Measure	**Naïve Bayes**	**SVM**	**KNN**	**MLP**	**Hoeffding**	**NBTree**
Accuracy	91.27	92.26	89.2	92.78	93.96	96.36
Kappa	0.93	1	0.88	1	1	1

RMSE	0.13	0	0.21	0.01	0.01	0.01
F Measure	0.98	1	0.97	1	1	1
Precsion	0.99	1	0.97	1	1	1
Recall	0.97	1	0.96	1	1	1
methews Correlation	0.93	1	0.88	1	1	1
Area_under_ROC	1	1	0.94	1	1	1
Area_under_PRC	1	1	0.96	1	1	1

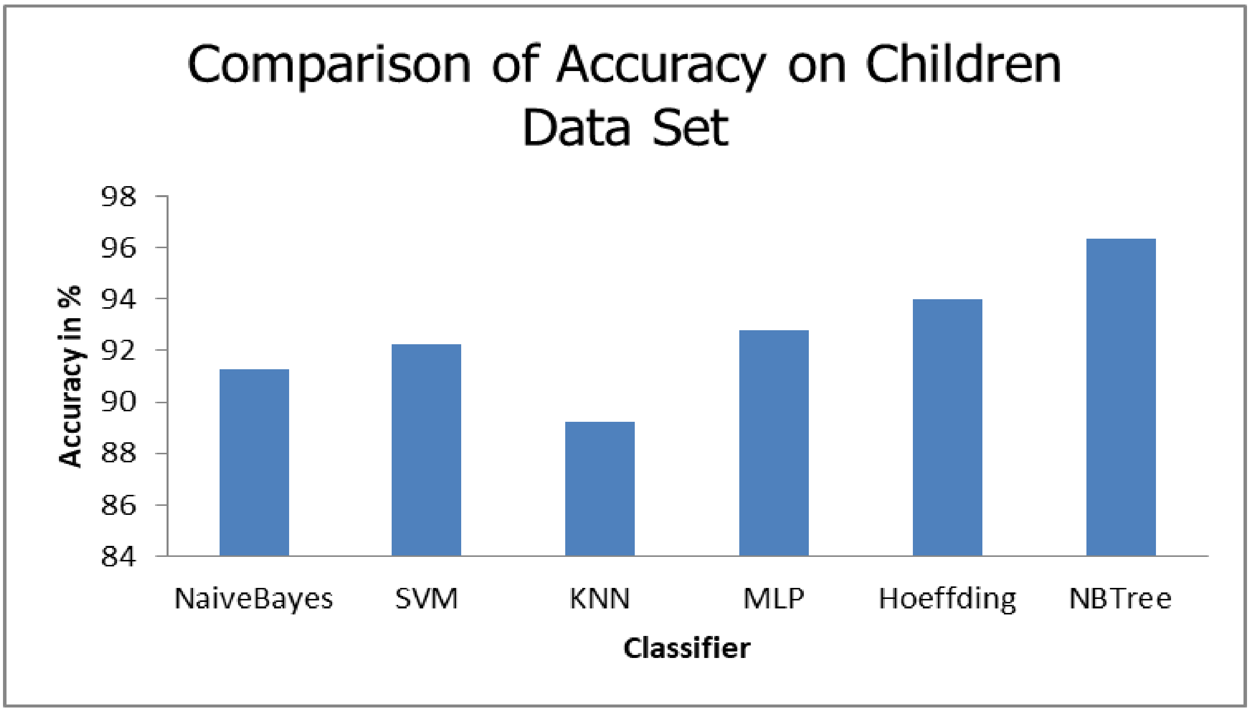

Fig.4 Accuracy of classifiers on Children Dataset

The accuracy of the NB Tree based classifier is the best among the other five classifiers on the children data set. Except KNN, Navie Bayes,SVM, MLP and Hoeffding have practically identical accuracy in terms of prediction. Perhaps, the performance of SVM, Hoeffding and NB Tree is identical in virtually all performance criteria except accuracy.

The experimental investigation to discover the optimum classifier to design the prediction system for all sorts of age groups demonstrates that Navie Bayes Tree (NB Tree) is more suited than widely used classifiers.

V. CONCULSION:

Autism disease and associated problems impact 10 to 15 of every 10,000 persons globally. Using historical data sets, several intelligent approaches have been developed to determine if a person has ASD or not. This research provides a classification method for ASD data based on the Naive Bayes Tree. Since NB Tree is straightforward and simple to comprehend, like a decision tree, it may be used by many types of healthcare practitioners. In addition, its classification accuracy improves as the size of the data set grows. The same classification approach may be used to identify individuals with ASD, regardless of their age group. After putting the approach into the autism detection tool, the performance has yet to be evaluated on an actual data set. In the future, its performance should also be compared to hybrid machine learning techniques

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